No. of Functional Features Included in the Solution

Date	28 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Functional Features Overview:

This document lists the functional features implemented in the Rising Waters: A Machine Learning Approach to Flood Prediction project. Each feature contributes to building an intelligent flood prediction system using Machine Learning and a Flask web application.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Data Collection	Collects historical flood and weather data from Kaggle	Dataset Module	Done	4
2	Data Preprocessing	Cleans data, handles missing values, feature scaling	Data Processing	Done	6
3	Data Visualization	Generates plots and statistical analysis of flood data	Visualization Module	Done	5
4	Machine Learning Model	Trains and evaluates Decision Tree, Random Forest, KNN, and XGBoost models	ML Module	Done	12
5	Flood Prediction	Predicts flood occurrence using the trained XGBoost model	Prediction Engine	Done	8
6	Flask Web Application	Provides a web interface for user interaction and prediction	Flask Application	Done	7
7	Prediction Result Display	Displays Flood Chance or No Flood Chance based on prediction	Result Module	Done	4
8	IBM Cloud Deployment	Deploys the application for online access	Deployment Module	Pending	4

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	7

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Home Page, Prediction Input & Result Pages
2	Backend / Logic	3	Data Preprocessing, ML Prediction, Flask Routing
3	Database / Storage	1	Dataset (CSV) and Saved Model (save/pkl)
4	API / Integration	1	Flask Integration with Machine Learning Model
5	Security / Authentication	0	Not Implemented in Current Version